Final Examination: Solutions and Grading Guide

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Notes for graders. As on the midterm: strategies must be complete plans; carry-through errors cost points only once; decimals to three places are acceptable wherever fractions are given. Where a part asks for an explanation, one or two correct sentences earn the explanation points; length is not rewarded.

Problem 1, Part I [20 points]

(a) Backward induction: at Q4, L takes ($10 > 8$); at Q3, H takes ($8 > 5$); at Q2, L takes ($6 > 3$); at Q1, H takes ($4 > 2$). The SPE is: H takes at Q1 and at Q3; L takes at Q2 and at Q4. The outcome is immediate dissolution with payoffs $(4, 1)$. The total value is 5, compared with 20 if the venture ran to the sale, so 15 (75% of the potential value) is lost.

Grading (5 points).

- 2 pts: correct backward-induction comparisons at all four nodes.
- 1 pt: complete strategies (including Take at Q3 and Q4). 1 pt: outcome $(4, 1)$. 1 pt: loss of 15.

(b) Each partner moves at two nodes with two actions each, so each has $2^2 = 4$ pure strategies. H's strategies (action at Q1, action at Q3): (T,T), (T,P), (P,T), (P,P).

Grading (4 points).

- 1 pt each for the two counts. 2 pts: H's list (including strategies such as (T,P) that specify actions at unreached nodes).

(c) Fix a Nash equilibrium (with mixed strategies, equivalently behavior strategies, since the game has perfect recall). In a Nash equilibrium, every action chosen with positive probability at a node reached with positive probability must be optimal there, given the opponent's strategy. Let k^* be the last node reached with positive probability, where "node 5" denotes the final sale.

- $k^* = 5$ is impossible: then L passes with positive probability at Q4, which is reached, but Take gives $10 > 8$.
- $k^* \in \{2, 3, 4\}$ is impossible: the mover at k^* takes with probability one (otherwise node $k^* + 1$ would be reached), so the opponent, who moves at $k^* - 1$ and passes there with positive probability, receives the Take payoff at k^*. This is strictly less than what she gets by taking herself at $k^* - 1$: for H, $2 < 4$ (at Q1) and $5 < 8$ (at Q3); for L, $3 < 6$ (at Q2). Shifting that probability to Take is a profitable deviation.

Hence $k^* = 1$: Q2 is reached with probability zero, i.e., H takes at Q1 with probability one. (Strategies at later nodes can differ across Nash equilibria, but the outcome $(4, 1)$ is the same in all of them.)

Grading (5 points).

- 1 pt: correct use of "optimal at reached nodes".
- 3 pts: the last-reached-node argument, including the payoff inequalities (2 pts if the student only treats pure strategies).
- 1 pt: conclusion and the remark that off-path behavior may differ.

(d) With final payoffs $(12-b,\ 8+b)$ and ties resolved in favor of Pass:

- Q4: L passes iff $8+b \geq 10$, i.e., $b \geq 2$.
- Q3 (given L passes at Q4): H passes iff $12-b \geq 8$, i.e., $b \leq 4$.
- Q2: L passes since $8+b \geq 10 > 6$. Q1: H passes since $12-b \geq 8 > 4$.

So full cooperation is the SPE iff $2 \leq b \leq 4$. H chooses the smallest such bonus, $b = 2$, and the payoffs are $(10, 10)$: H gains 6 relative to the SPE without a contract, and the full value of 20 is realized.

With $b = 4.5$ the final payoffs are $(7.5,\ 12.5)$. L would pass at Q4, but then H prefers to take at Q3 ($8 > 7.5$). Anticipating this, L takes at Q2 ($6 > 3$), and H takes at Q1 ($4 > 2$): the venture again unravels to $(4, 1)$. A bonus that is too generous destroys the *payer's* own incentive to continue.

Grading (6 points).

- 2 pts: condition $b \geq 2$ from Q4. 2 pts: condition $b \leq 4$ from Q3.
- 1 pt: $b = 2$ with payoffs $(10, 10)$. 1 pt: $b = 4.5$ unravels, with the Q3 comparison.

Problem 1, Part II [30 points]

(e) L accepts any offer giving it at least 0 (it is indifferent at 0 and accepts). So H proposes $(1, 0)$ and L accepts. SPE strategies: H proposes $(1, 0)$; L accepts every offer. Outcome $(1, 0)$.

Grading (4 points).

- 2 pts: L's acceptance rule. 2 pts: H's offer and the outcome.

(f) Work backward (shares measured in the value at the time of the offer).

- Round 3 (H proposes): H takes 1, L accepts 0.
- Round 2 (L proposes): H accepts iff offered at least $\delta_H \cdot 1 = 0.9$. L offers H 0.9 and keeps 0.1.
- Round 1 (H proposes): L accepts iff offered at least $\delta_L \cdot 0.1 = 0.08$. H offers L 0.08 and keeps 0.92.

Agreement is immediate with split $(0.92,\ 0.08)$.

Grading (6 points).

- 1 pt: round 3. 2 pts: round 2 (offer 0.9 to H). 2 pts: round 1 (offer 0.08 to L). 1 pt: immediate agreement.

(g) Let x be H's share when H proposes and y be L's share when L proposes. In a stationary SPE each responder is exactly indifferent between accepting and waiting one round to propose:

$$1 - x = \delta_L y, \qquad 1 - y = \delta_H x.$$

Solving, $x = \dfrac{1-\delta_L}{1-\delta_H\delta_L} = \dfrac{0.2}{0.28} = \dfrac{5}{7}$ and $y = \dfrac{1-\delta_H}{1-\delta_H\delta_L} = \dfrac{0.1}{0.28} = \dfrac{5}{14}$.

Equilibrium strategies (in every round, after every history):

- H proposes $(\frac{5}{7},\ \frac{2}{7})$ and accepts an offer iff it gives her at least $\delta_H x = \frac{9}{14}$.

- L proposes $(\frac{9}{14}, \frac{5}{14})$ and accepts an offer iff it gives it at least $\delta_L y = \frac{2}{7}$.

Outcome: H's first offer is accepted immediately; H gets $\frac{5}{7} \approx 0.714$ and L gets $\frac{2}{7} \approx 0.286$. (This is the unique SPE of the Rubinstein game.)

Grading (8 points).

- 2 pts: stationarity (indifference) equations. 2 pts: $x = 5/7$ (and $y = 5/14$).
- 2 pts: complete strategies with acceptance thresholds $9/14$ and $2/7$.
- 2 pts: immediate agreement and final split. Students who only quote the formula without the equations: at most 5 of 8.

(h) With L proposing first, L gets $y = \frac{5}{14} \approx 0.357$ and H gets $\frac{9}{14} \approx 0.643$. Moving first is worth something to each partner (L: $0.286 \to 0.357$; H: $0.643 \to 0.714$), but patience matters more: the more patient Harbor Authority gets more than half of the lease even when it proposes second.

Grading (6 points).

- 3 pts: shares $5/14$ and $9/14$.
- 3 pts: comment on both first-mover advantage and patience (1.5 each).

(i)

$$x(\Delta) = \frac{1 - e^{-r_L \Delta}}{1 - e^{-(r_H + r_L)\Delta}} \xrightarrow[\Delta \to 0]{} \frac{r_L}{r_H + r_L} = \frac{0.12}{0.16} = \frac{3}{4}$$

(by a first-order expansion $1 - e^{-z} \approx z$, or L'Hôpital). As offers can be made more and more quickly, the first-mover advantage disappears and the split depends only on relative impatience: L discounts three times as fast as H, so H gets three quarters. (For comparison, with $\Delta = 1$ month H's share is about 0.765.)

Grading (6 points).

- 2 pts: $x(\Delta)$ in terms of r's. 2 pts: limit $3/4$. 2 pts: interpretation.

Problem 2 [50 points]

(a) Arcadia maximizes $(90 - q_A - q_B)q_A$, so $q_A = \dfrac{90 - q_B}{2}$; Brightwave maximizes $(84 - q_A - q_B)q_B$, so $q_B = \dfrac{84 - q_A}{2}$ (both truncated at 0). Solving: $q_A^* = 32$, $q_B^* = 26$, $P = 100 - 58 = 42$. Profits: $\pi_A = (42 - 10) \cdot 32 = 1024$ and $\pi_B = (42 - 16) \cdot 26 = 676$.

Grading (8 points).

- 2 pts each: the two best responses. 2 pts: quantities. 1 pt: price. 1 pt: profits.

(b) Let $BR_A(q_B) = \max\{0, (90 - q_B)/2\}$ and $BR_B(q_A) = \max\{0, (84 - q_A)/2\}$. Each is decreasing with slope $-\frac{1}{2}$, so if the opponent's surviving quantities form an interval $[\ell, u]$, the quantities that are best responses to some belief over $[\ell, u]$ form the interval $[BR(u), BR(\ell)]$.

Round	Arcadia q_A	Brightwave q_B
1	$[0,\ 45]$	$[0,\ 42]$
2	$[BR_A(42), BR_A(0)] = [24,\ 45]$	$[BR_B(45), BR_B(0)] = [19.5,\ 42]$
3	$[BR_A(42), BR_A(19.5)] = [24,\ 35.25]$	$[BR_B(45), BR_B(24)] = [19.5,\ 30]$

In round 1, quantities above the monopoly outputs 45 and 42 are never best responses. Each subsequent round the width of one firm's interval is half the width of the other firm's interval in the previous round, so both widths shrink geometrically to zero (round 4: $[30,\ 35.25]$ and $[24.375,\ 30]$). The bounds converge to the fixed point of the best responses, $(32, 26)$, which is therefore the unique rationalizable profile.

Grading (8 points).

- 2 pts: round 1. 2 pts: round 2. 2 pts: round 3.
- 2 pts: convergence argument (slope $1/2$ contraction or explicit halving of widths).

(c) Brightwave's strategy is its best-response function $q_B(q_A) = \max\{0, (84 - q_A)/2\}$. Arcadia maximizes

$$\Big(100 - q_A - \frac{84 - q_A}{2} - 10\Big) q_A = \Big(48 - \frac{q_A}{2}\Big) q_A \quad \Longrightarrow \quad q_A = 48.$$

Then $q_B = 18$, $P = 34$, $\pi_A = 24 \cdot 48 = 1152$ and $\pi_B = 18 \cdot 18 = 324$. Arcadia gains 128 relative to Cournot (first-mover advantage: committing to a large quantity makes Brightwave cut back), Brightwave loses 352, and consumers gain because output rises from 58 to 66 (consumer surplus $1682 \to 2178$).

Grading (8 points).

- 2 pts: Brightwave's response function (as a strategy). 2 pts: $q_A = 48$.
- 2 pts: q_B, price and profits. 2 pts: comparison with Cournot.

(d) The unique equilibrium is $p_A = p_B = 16$. Arcadia sells $Q = 84$ and earns $(16 - 10) \cdot 84 = 504$; Brightwave earns 0.

It is an equilibrium: if Arcadia raised its price it would lose all customers; lowering it only reduces its margin. Brightwave earns 0 at any price ≥ 16 (it either sells nothing or sells at cost).

It is the only one: (i) if $p_A = p_B = p > 16$, Brightwave could undercut slightly and earn a positive profit; (ii) if $16 \leq p_B < p_A$, Arcadia earns 0 but could earn $(p_B - 10)(100 - p_B) > 0$ by matching p_B; (iii) if $p_A < p_B$, Arcadia sells to everyone and would move its price to $\min\{p_B, 55\}$ (55 is its monopoly price) unless it is already there; if $p_A = 55 < p_B$, Brightwave could undercut Arcadia at a price in $(16, 55)$ and earn a positive profit. Prices below 16 for Brightwave are excluded by assumption.

Compared with Cournot, Arcadia earns much less (504 vs. 1024): it can only price at its rival's cost (limit pricing). Consumer surplus rises from $58^2/2 = 1682$ to $84^2/2 = 3528$.

Grading (8 points).

- 2 pts: equilibrium prices. 1 pt: profits.
- 3 pts: verification plus uniqueness argument (the case $p > 16$ is the key one: 2 pts).
- 2 pts: comparison of profit and consumer surplus with Cournot.

(e) With both costs equal to 10:

- Collusion: monopoly output 45 at price 55; each firm produces 22.5 and earns $\pi^m = 1012.5$.

- Deviation: the best response to 22.5 is 33.75, the price is 43.75, and the profit is $\pi^d = 33.75^2 = 1139.0625$.
- Punishment: Cournot, $q = 30$ each, price 40, profit $\pi^c = 900$. Cournot reversion repeats a stage Nash equilibrium, so it is credible.

Collusion is sustainable iff $\dfrac{\pi^m}{1-\delta} \geq \pi^d + \dfrac{\delta\pi^c}{1-\delta}$, i.e.,

$$\delta \;\geq\; \frac{\pi^d - \pi^m}{\pi^d - \pi^c} = \frac{126.5625}{239.0625} = \frac{9}{17} \approx 0.529.$$

With $\delta = e^{-r/4}$: $e^{-r/4} \geq 9/17 \iff r \leq 4\ln(17/9) \approx 2.54$, i.e., a continuously compounded rate of about 254% per year. At any plausible interest rate, quarterly interaction makes full collusion sustainable.

Grading (10 points).

- 1 pt each: π^m, π^d (with the deviation quantity), π^c (3 pts).
- 1 pt: credibility of the punishment.
- 2 pts: incentive constraint. 2 pts: $\delta \geq 9/17$.
- 2 pts: $r \leq 4\ln(17/9) \approx 2.54$.

(f) With quota q: $\pi^{col}(q) = (90-2q)q$, the best deviation is $(90-q)/2$ with $\pi^{dev}(q) = \left(\frac{90-q}{2}\right)^2$, and $\pi^c = 900$. Then

$$\pi^{dev} - \pi^{col} = \frac{(90-q)^2 - 4q(90-2q)}{4} = \frac{9(30-q)^2}{4},$$
$$\pi^{dev} - \pi^{c} = \frac{(90-q)^2 - 60^2}{4} = \frac{(30-q)(150-q)}{4},$$

so the smallest sustaining discount factor is

$$\delta(q) = \frac{\pi^{dev} - \pi^{col}}{\pi^{dev} - \pi^{c}} = \frac{9(30-q)}{150-q}.$$

Check: $\delta(22.5) = \frac{9}{17}$ as in (e). At $q = 25$: $\delta(25) = \frac{45}{125} = \frac{9}{25} = 0.36$.

$\delta(q)$ is decreasing in q (less ambitious agreements are easier to sustain), while per-firm profit $(90-2q)q$ is decreasing in q on $[22.5, 30]$. With $\delta = \frac{1}{2}$, the constraint $\frac{9(30-q)}{150-q} \leq \frac{1}{2}$ gives $18(30-q) \leq 150-q$, i.e., $q \geq \frac{390}{17} \approx 22.94$. The most profitable sustainable agreement is $q = \frac{390}{17}$ per firm, with profit $\frac{292500}{289} \approx 1012.11$ each: only about 0.39 below the full-collusion profit of 1012.5.

Grading (8 points).

- 3 pts: derivation of $\delta(q)$ (2 pts for the two differences, 1 pt for simplifying).
- 1 pt: $\delta(25) = 9/25$.
- 2 pts: solve for $q \geq 390/17$ with the monotonicity argument.
- 2 pts: profit ≈ 1012.11 and comparison with full collusion.